

Multi-Agent Platform for HNWI News Monitoring and Actionable Insights

CS480 Project Experience



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PRODUCT DEMO

Overview

PROBLEM STATEMENT

Private banking advisors have to process lots of online information when monitoring private banking clients and conducting background research on potential clients, having to internalise news summaries and form value judgements on their own, leading to inefficiencies or cognitive biases

PROJECT SCOPE

Private banking advisors' responsibilities:

1. Conducting background research on potential private banking clients
2. Ensuring current clients are bank-compliant/in good public standing

Accomplished via accessing publicly accessible information on high net worth individuals/entities (News sources, financial blogs, auditor reports)

SOLUTION

Client News Screener: efficient, reliable assistant

Multi-agentic, human-in-the-loop conversational AI that can retrieve online information on high net worth individuals, summarize online sources, and advise private bank advisors on the individual's suitability as a private banking client or to assess current client compliance

Competitor Analysis



Aggregates real-time insights from public data, uses NLP filters to put on dashboards, alerting stakeholders



Predictive intelligence on private companies using rich data signals. Uses purpose-built AI agents to surface relevant insights

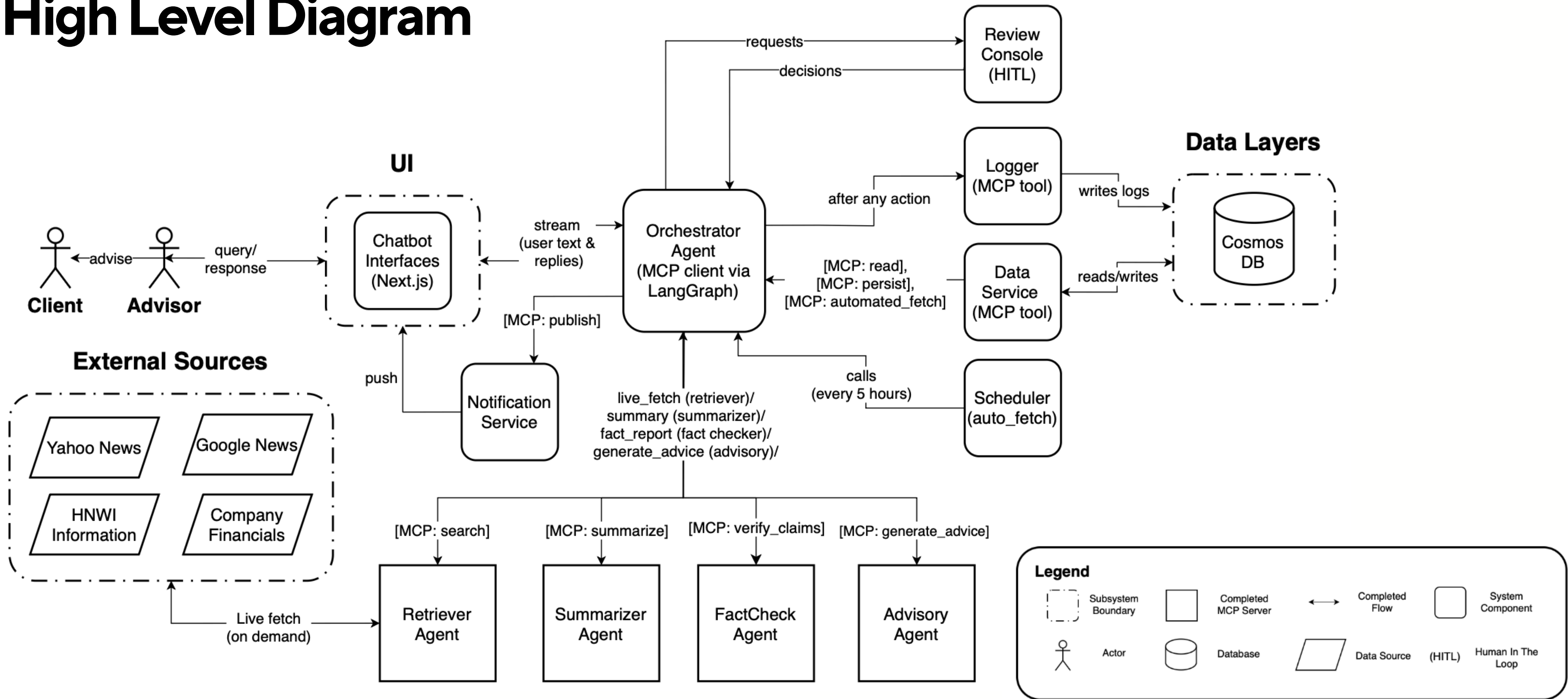


AI market intelligence search platform tapping into premium content, turns fragmented data into insight

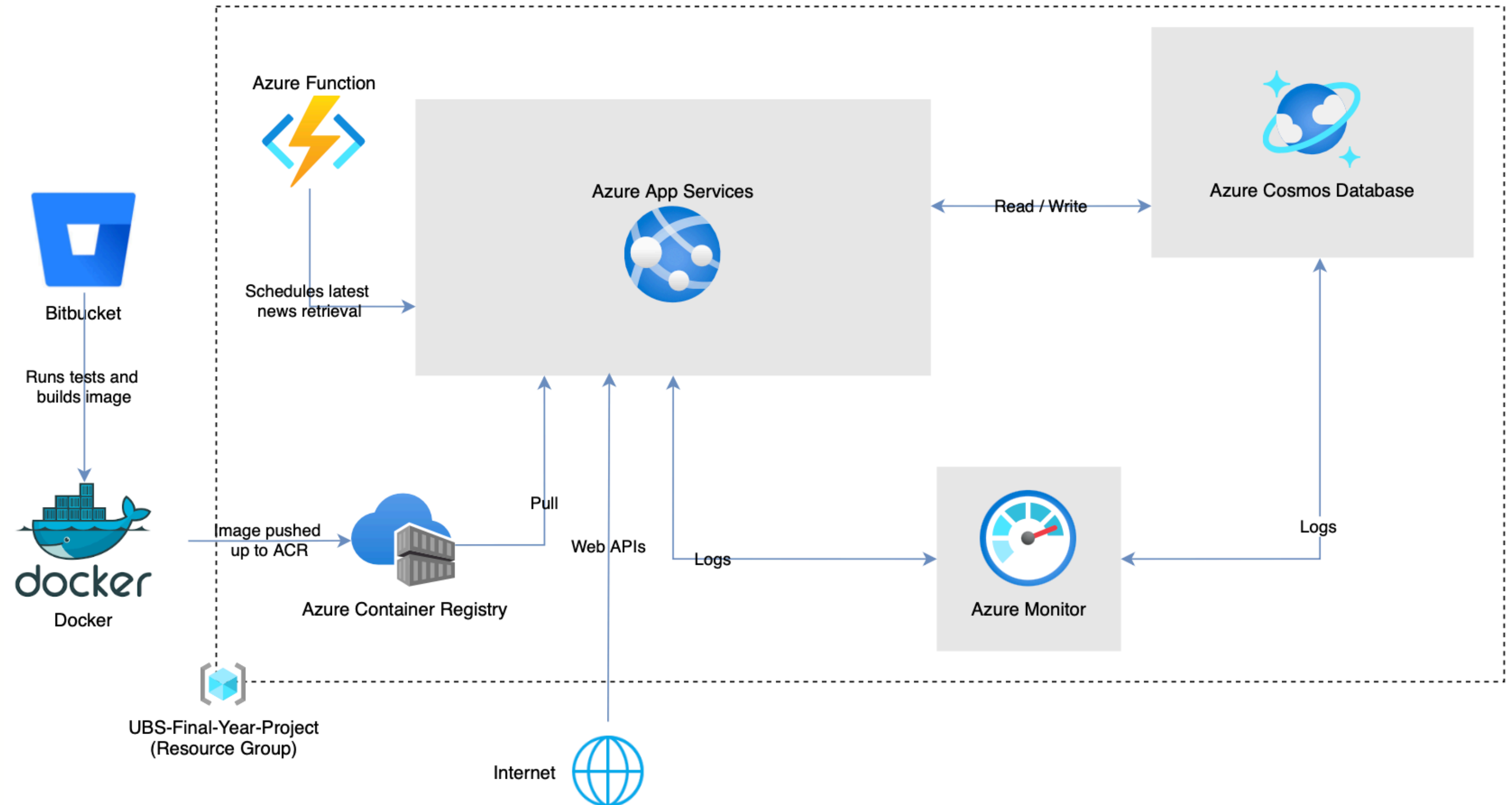
PRODUCT DEMO

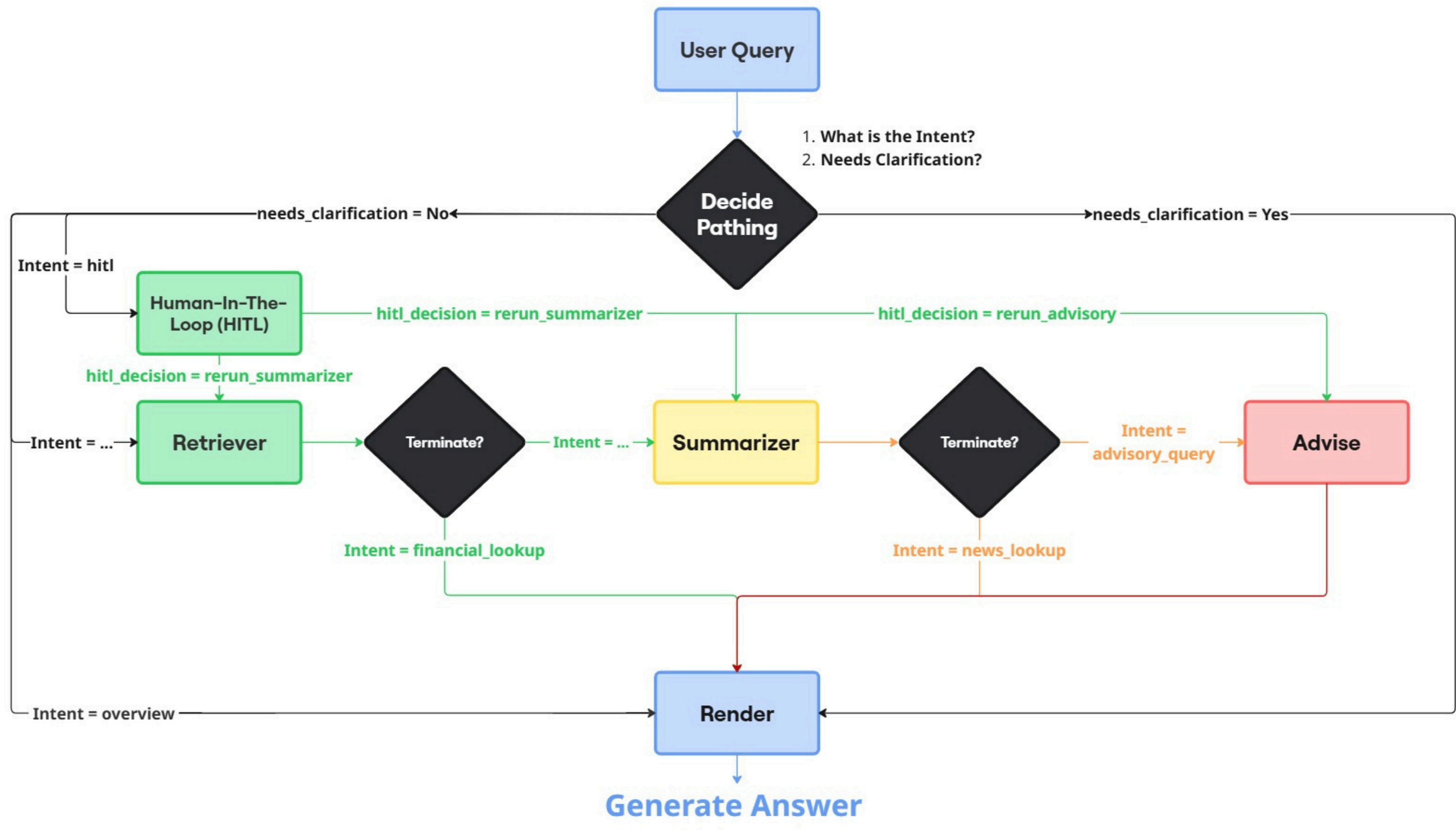
Architecture

High Level Diagram



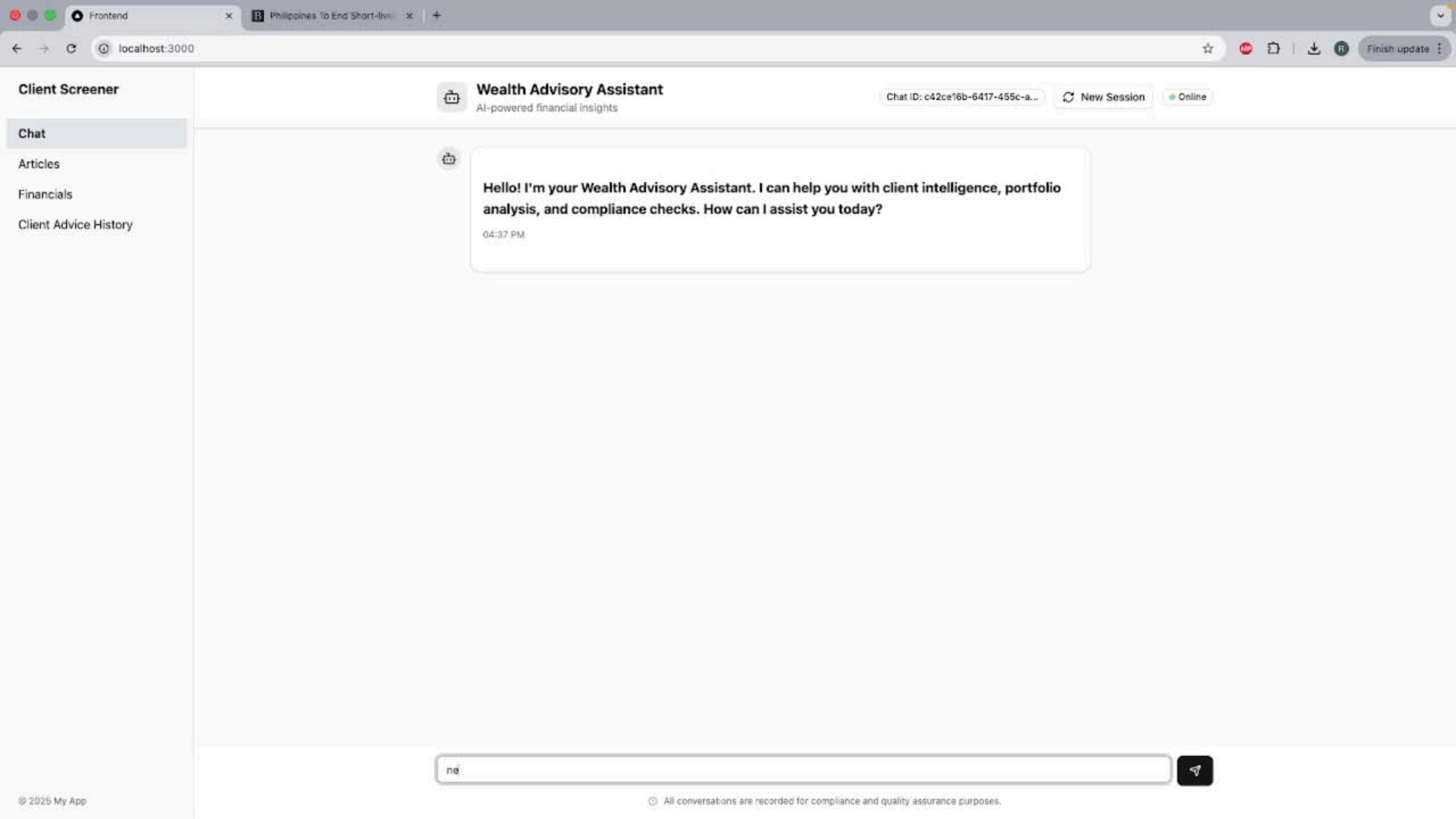
Infrastructure Diagram





PRODUCT DEMO

DEMO



Client Screener

Chat

Articles

Financials

Client Advice History



Wealth Advisory Assistant

AI-powered financial insights

Chat ID: c42ce16b-6417-455c-a...

New Session

Online



Hello! I'm your Wealth Advisory Assistant. I can help you with client intelligence, portfolio analysis, and compliance checks. How can I assist you today?

04:37 PM

na



PRODUCT DEMO

Evaluation

Summarization Quality

Context

- Articles are freshly scraped → no human reference for summaries
- We evaluate reference-free metrics that align with advisor needs

FactCC

Consistency of summary statements with the source

BERTScore

Meaning overlap with the article

n-gram Ratio

Non-repetitiveness of the summary text

Flesch-Kincaid Grade

Reading grade level

Summary Length

Absolute tokens/chars in the summary

Compression Ratio

compactness

Summarization Quality - Results

model	factcc_score	bertscore	2-gram_ratio	fk_grade	summary_length	compression_ratio
deepseek-chat-v3.1	0.754	0.895	1.000	13.836	74.667	0.658
gemini-2.5-flash	0.044	0.863	0.996	12.485	69.000	0.381
llama-4-maverick	0.066	0.852	1.000	11.847	58.333	0.338
mai-ds-r1	0.346	0.852	1.000	12.936	65.667	0.374
qwen3-4b	0.307	0.850	1.000	13.432	61.000	0.350

deepseek best matched our priorities

- high factual consistency
- strong semantic alignment

Advisory Generation - Confident AI Details

Test Case Details

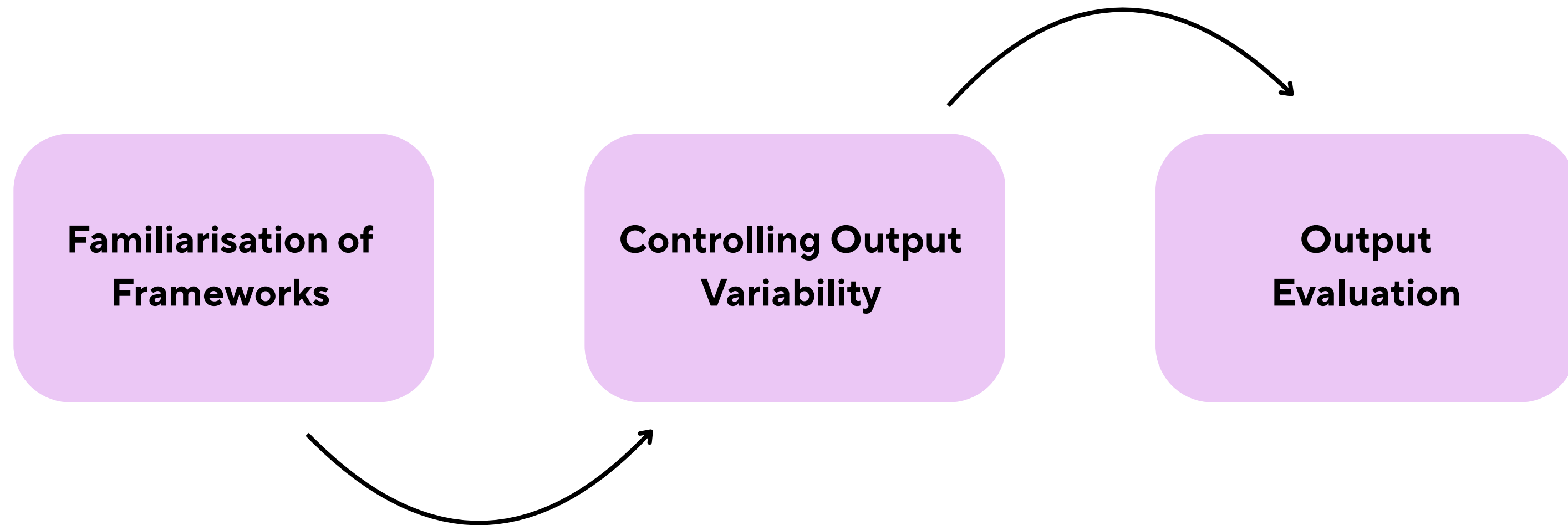
- 3 main metrics used
 - Advice Completeness
 - Advice Correctness
 - Answer Relevancy
- Scores assigned from 0 to 1
- Used stronger model (GPT-4o-mini) for comparing expected and actual advisory output
- Provides qualitative feedback on actual output to improve advisory system prompt

Test Case Details		✕
Test Case ID	d86f323f-e965-4bba-be50-deb3b13612aa	
Status	Success	
Run Duration	24.5s	
Metrics	✓ Advice Completeness [GEval] ✓ Advice Correctness [GEval] ✓ Answer Relevancy	
Metrics Data	Test Case Parameters	Additional Info
Metric Name	Advice Completeness [GEval]	
Status	Success	
Evaluation Model	openrouter/openai/gpt-4o-mini (('openai/gpt-4o-mini', 'openrouter', None, None))	
Score	0.88	Threshold 0.5
Reason	The actual output closely aligns with the expected output, accurately reflecting the key information from the input prompt. It correctly identifies that no information is available regarding Lisa Su's net worth, investment portfolio, or investment activity, which is consistent with the articles reviewed. The associations section is comprehensive, listing relevant companies and individuals with appropriate citations. The risk assessment accurately notes the absence of controversies and assigns low risk ratings based on the positive portrayal of Lisa Su in the articles. The suitability analysis provides a strong justification for the overall rating and service usage likelihood, though the likelihood rating is slightly lower than expected, indicating a minor deviation from the expected output.	

PRODUCT DEMO

AI Journey

Our AI Journey



PRODUCT DEMO

Conclusion

Current and Future Works

What we've achieved

- Conversational AI with full core functionalities
- Robust Orchestrator With Dynamic LangGraph Workflow
- Fully stateful sessions with memory and context reuse
- Modular MCP servers with clean JSON-RPC interfaces (industry-standard protocol)
- Reproducible LLM-as-Judge scoring pipeline for evaluation
- Clean and intuitive UI
- Full deployment with CI

Current and Future Works

Our Long-Term Vision for This Platform

- Adopt locally fine-tuned LLMs
 - Data Protection and Performance
- Transition fully into a microservice architecture
 - Modularity and scalability
- Strengthen CI/CD and observability

There are lots of room to grow!

Thank you!

QnA